WI-FI SECURITY ASSESSMENT OFFICE PROJECT

3MTT FELLOW / NEXT GENERATION CAPSTONE PROJECT

BY

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Assessment Scope: Authorized assessment of the project participant's Airtel 4G/5G Wi-Fi router and connected laboratory devices, using the supplied screenshots and recorded assessment outputs.

Executive Summary

This final assessment consolidates the Wi-Fi security assessment performed against an authorized Airtel 4G/5G Wi-Fi router in a controlled laboratory environment. The assessment covered network identification, service discovery, HTTP/HTTPS management exposure, TLS certificate and cipher assessment, wireless authentication review, WPS assessment, remediation, and post-remediation verification.

The principal confirmed security finding was WPS being enabled on both the 2.4 GHz and 5 GHz wireless networks. The supplied evidence shows that WPS was subsequently disabled on both bands and that post-remediation screenshots confirmed the disabled state. Connectivity was also tested after remediation, and a post-remediation Nmap scan was captured.

The router was observed using WPA2-PSK authentication on both wireless bands. The interface also exposed WPA3-PSK and WPA2/WPA3-PSK options. WPA2-PSK is therefore recorded as a hardening opportunity rather than a confirmed vulnerability. The exact 802.11 wireless cipher was not independently verified because the Kali virtual machine reported no wireless interface through iw dev and iwconfig.

1. Assessment Objectives

a) Establish the network and device baseline for the authorized router.
b) Identify exposed network services and management interfaces.
c) Review HTTP/HTTPS security headers and TLS configuration.
d) Review wireless authentication options on 2.4 GHz and 5 GHz.
e) Assess the state of WPS on both wireless bands.
f) Document security findings and assess their practical risk.
g) Apply an authorized remediation to the confirmed WPS finding.
h) Verify the remediation and preserve evidence of the post-remediation state.

2. Scope and Authorization

The assessment was limited to the participant's Airtel router and devices in the authorized laboratory environment. No third-party systems were intentionally targeted. The project was performed as a cybersecurity learning/capstone exercise with the objective of assessing and improving the security posture of the test environment.

The evidence package includes an ethical-scope diagram and network diagram supplied by the participant.

3. Environment and Evidence

Component	Observed/Used	Evidence
Router	Airtel 4G/5G Wi-Fi router; management IP 192.168.1.1	Airtel Router.png
Assessment host	Kali Linux virtual machine	Kali IP.png
Client	Windows laptop	Laptop IP.png / Verify Windows laptop.png
Mobile client	Participant phone	My phone.jpeg
Network	192.168.1.0/24 LAN observed during assessment	Network Digram.drawio.png
Tools	Nmap, curl, iw, iwconfig; router web interface	Command screenshots throughout package

4. Methodology

The project was executed phase-by-phase. Evidence was captured at each major stage so that the final assessment could demonstrate a clear before/after security story.

1. Network and device baseline collection.
2. TCP service and version discovery against 192.168.1.1.
3. HTTP/HTTPS response inspection.
4. TLS certificate and supported cipher assessment.
5. Wireless authentication configuration review.
6. WPS baseline assessment on 2.4 GHz and 5 GHz.
7. Assessment of Kali wireless-interface capability and documentation of limitations.
8. WPS remediation through the router management interface.
9. Post-remediation WPS verification and connectivity verification.
10. Post-remediation Nmap verification and final risk review.

5. Network and Service Discovery Results

The supplied Nmap evidence identified the router at 192.168.1.1 as reachable. The captured port discovery showed TCP/53, TCP/80 and TCP/443 open, with 997 other TCP ports reported closed in the scan.

Service/version detection identified dnsmasq 2.90 on TCP/53. TCP/80 returned an HTTP response containing several security-related headers, while TCP/443 was available for HTTPS management.

Evidence: Port Discovery.png

Evidence: Service version detection.png

Evidence: HTTP Response.png

Evidence: HTTPS Confirm.png

Evidence: HTTPS Response.png

6. TLS and Certificate Assessment

The TLS evidence shows TLS 1.2 and TLS 1.3 support. The supplied Nmap ssl-enum-ciphers output reported an overall least-strength rating of A and included AES-GCM and ChaCha20-Poly1305 suites among the TLS 1.2/1.3 results. The certificate evidence also shows an RSA public key of 4096 bits and SHA-256 with RSA encryption.

The assessment did identify a warning in the TLS enumeration output concerning the key-exchange strength (secp256r1) relative to the certificate key. This is recorded as an observation for technical review rather than a confirmed exploitable vulnerability.

Primary evidence: TLS Certificate.png and the captured ssl-enum-ciphers terminal output.

7. Wireless Authentication Assessment

The router management interface showed WPA2-PSK as the selected authentication method on both 2.4 GHz and 5 GHz. The available authentication options shown in the router interface included OPEN, WPA2-PSK, WPA/WPA2-PSK, WPA3-PSK, and WPA2/WPA3-PSK.

The presence of WPA3-capable options provides a future hardening opportunity. The project did not change the authentication method because the primary confirmed finding selected for remediation was WPS.

8. WPS Baseline Finding

The 2.4 GHz and 5 GHz WPS baseline screenshots show the WPS Enable controls in the enabled state. This was treated as the principal confirmed security finding because WPS can increase the attack surface of a wireless network.

Finding ID	Finding	Affected Bands	Status
WIFI-01	WPS enabled	2.4 GHz and 5 GHz	Remediated
WIFI-02	WPA2-PSK retained while WPA3 is available	2.4 GHz and 5 GHz	Hardening opportunity
OBS-01	Exact wireless cipher not independently verified	Wireless	Assessment limitation
OBS-02	TCP/80 and TCP/443 management services exposed	Router	Review/monitor
POS-01	TLS 1.2/1.3 and Nmap least-strength A	HTTPS	Positive control

9. Remediation

The WPS setting was disabled on the router for both wireless bands. The remediation evidence shows the WPS Enable toggle in the disabled state for 2.4 GHz and 5 GHz.

Evidence: Phase-9-01-2.4GHz-WPS-After.png and Phase-9-02-5GHz-WPS-After.png.

10. Post-Remediation Verification

The post-remediation evidence set includes dedicated verification screenshots for both 2.4 GHz and 5 GHz WPS, a Kali connectivity check, and a post-remediation Nmap scan. This provides evidence that the remediation was performed and that the assessment environment remained operational.

- Phase-10-01-2.4GHz-WPS-Verified.png
- Phase-10-02-5GHz-WPS-Verified.png
- Phase-10-03-Kali-Connectivity-After.png
- Phase-10-04-Post-Remediation-Nmap.png

11. Wireless Cipher Assessment Limitation

The Kali VM did not expose a physical wireless interface. The supplied evidence from iw dev and iwconfig shows that the available interfaces were lo and eth0 and that both reported no wireless extensions. Consequently, the exact router wireless cipher (for example AES/CCMP, TKIP, or a mixed mode) could not be independently enumerated from Kali.

This is an evidence limitation, not a finding that the router is using a weak cipher. For a future assessment, a supported physical Wi-Fi adapter with monitor-mode capability could be connected to the assessment host, subject to authorization and compatibility.

12. Risk Assessment

Risk ratings below are scoped to the evidence and controls actually assessed in this project.

ID	Severity	Description	Treatment	Residual Status
WIFI-01	Medium	WPS enabled on both bands	Disable WPS	Remediated
WIFI-02	Low	WPA3 available but WPA2-PSK retained	Evaluate WPA3 compatibility	Open hardening opportunity
OBS-01	Informational	Cipher not independently verified	Repeat with physical Wi-Fi adapter	Open limitation
OBS-02	Informational	Web management ports exposed	Restrict management access where feasible	Review

Final risk position: no remaining confirmed medium, high, or critical findings were identified within the documented scope after the WPS remediation.

13. Security Recommendations

a) Keep WPS disabled on both wireless bands.

b) Evaluate WPA3-PSK or WPA2/WPA3-PSK for compatible client devices.

c) Use a long, unique Wi-Fi security key and avoid reusing passwords.

d) Use a unique router administrator password and protect management credentials.

e) Keep router firmware updated according to the vendor/carrier maintenance process.
f) Restrict router management access to trusted administration devices where the platform permits it.
g) Use a separate guest network for untrusted or visitor devices when available.
h) Review the connected-device list periodically and investigate unknown clients.
i) Disable unnecessary services or management interfaces where the router permits.
j) Repeat the security assessment after major firmware or configuration changes.

14. Assessment Limitations

a) The project used an Airtel router management interface whose available settings did not expose a direct wireless cipher field in the supplied screenshots.
b) The Kali assessment host was virtualized and did not have a usable physical Wi-Fi interface.
c) The assessment was limited to the authorized laboratory environment and the tests documented in the evidence package.
d) The assessment does not constitute a guarantee that the router contains no other vulnerabilities outside the tested scope.

15. Conclusion

The Wi-Fi security assessment successfully demonstrated a complete cybersecurity assessment lifecycle: establish a baseline, discover exposed services, assess security controls, identify a confirmed finding, apply remediation, and verify the result. The principal WPS finding was remediated on both 2.4 GHz and 5 GHz. The remaining observations are documented as hardening opportunities, informational items, or assessment limitations rather than overstated as vulnerabilities.

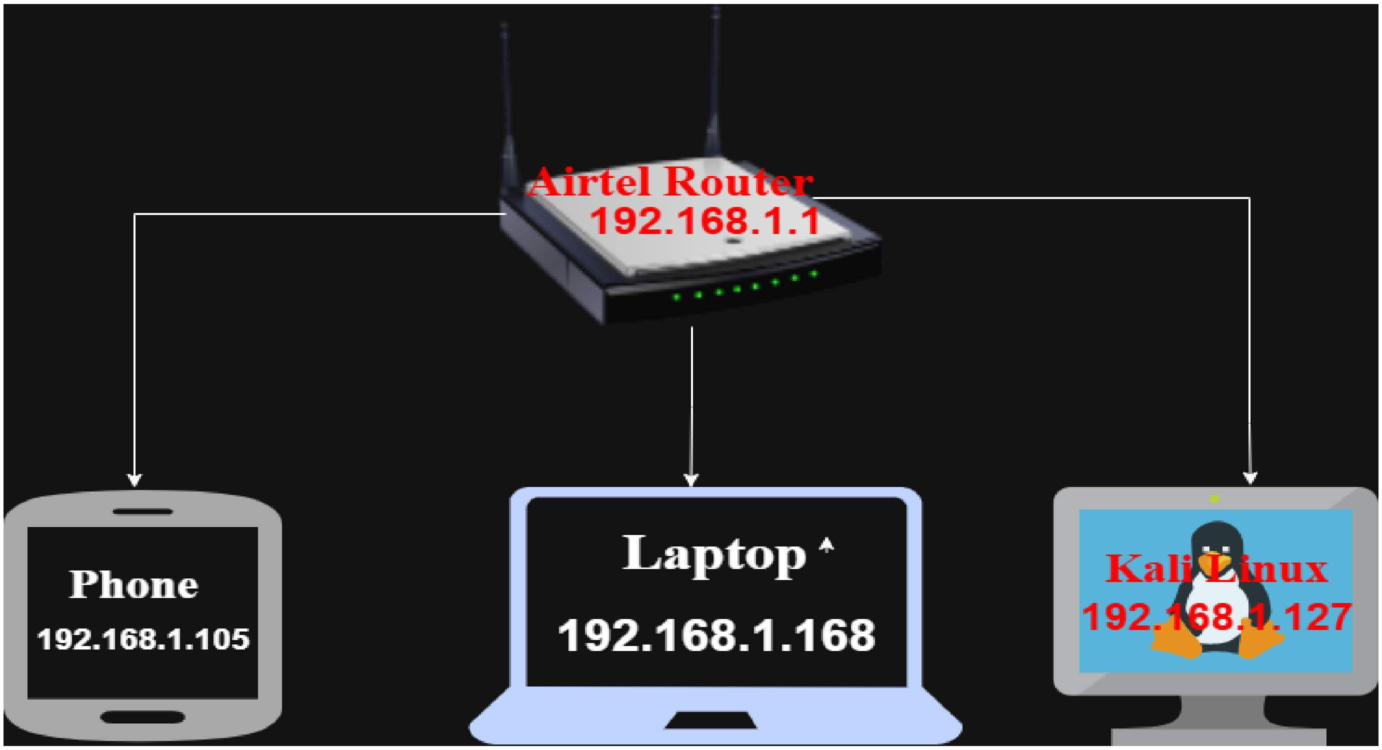
The supplied evidence package provides a traceable record of the assessment and supports the project's final presentation, defense, and review.

Appendix A - Evidence Register

- 01_Project_Setup_and_Scope — Airtel Router.png
- 01_Project_Setup_and_Scope — Ethical Scope Digram.drawio.png
- 01_Project_Setup_and_Scope — Network Digram.drawio.png
- 01_Project_Setup_and_Scope — Kali IP.png
- 01_Project_Setup_and_Scope — Laptop IP.png
- 01_Project_Setup_and_Scope — Verify Windows laptop.png
- 01_Project_Setup_and_Scope — My phone.jpeg
- 01_Project_Setup_and_Scope — Network Connectivity.png
- 01_Project_Setup_and_Scope — Test Internet network connectivity.png
- 02_Discovery_and_Service_Assessment — Port Discovery.png
- 02_Discovery_and_Service_Assessment — Service version detection.png
- 02_Discovery_and_Service_Assessment — HTTP Response.png
- 02_Discovery_and_Service_Assessment — HTTPS Confirm.png
- 02_Discovery_and_Service_Assessment — HTTPS Response.png
- 03_TLS_and_Certificate_Assessment — TLS Certificate.png
- 04_Wireless_Authentication_Assessment — Phase-5-01-2.4GHz-Authentication.png
- 04_Wireless_Authentication_Assessment — Phase-5-02-2.4GHz-Authentication-Options.png
- 04_Wireless_Authentication_Assessment — Phase-5-03-5GHz-Authentication.png
- 04_Wireless_Authentication_Assessment — Phase-5-04-5GHz-Authentication-Option.png
- 05_WPS_Baseline_Before_Remediation — Phase-6-01-2.4GHz-WPS-Before.png
- 05_WPS_Baseline_Before_Remediation — Phase-6-02-5GHz-WPS-Before.png
- 06_Wireless_Interface_Limitation — Phase-7-03-Kali-Wireless-Interface.png
- 06_Wireless_Interface_Limitation — Phase-7-04-IWCONFIG.png
- 07_Remediation_WPS_Disabled — Phase-9-01-2.4GHz-WPS-After.png
- 07_Remediation_WPS_Disabled — Phase-9-02-5GHz-WPS-After.png
- 08_Post_Remediation_Verification — Phase-10-01-2.4GHz-WPS-Verified.png
- 08_Post_Remediation_Verification — Phase-10-02-5GHz-WPS-Verified.png
- 08_Post_Remediation_Verification — Phase-10-03-Kali-Connectivity-After.png
- 08_Post_Remediation_Verification — Phase-10-04-Post-Remediation-Nmap.png
- 09_Unclassified_Reference — Screenshot (651).png

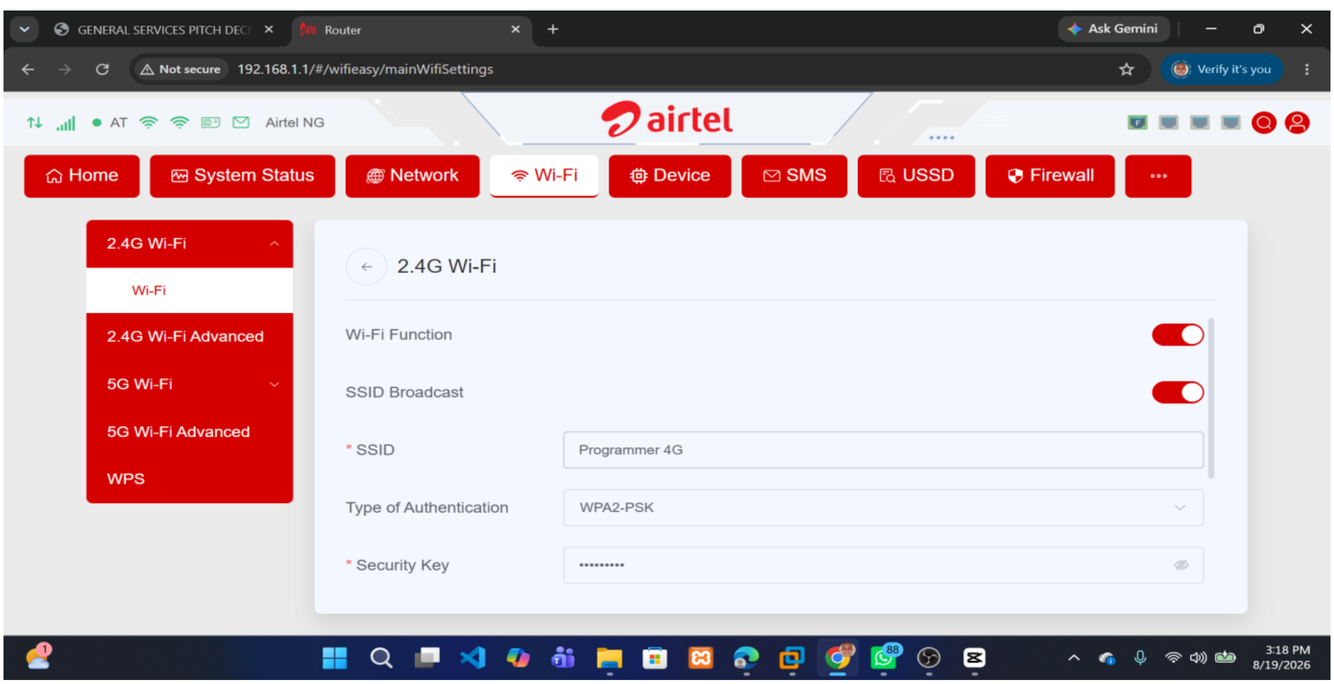
Appendix B - Selected Visual Evidence

Network architecture



Network Digram.drawio.png

2.4 GHz authentication

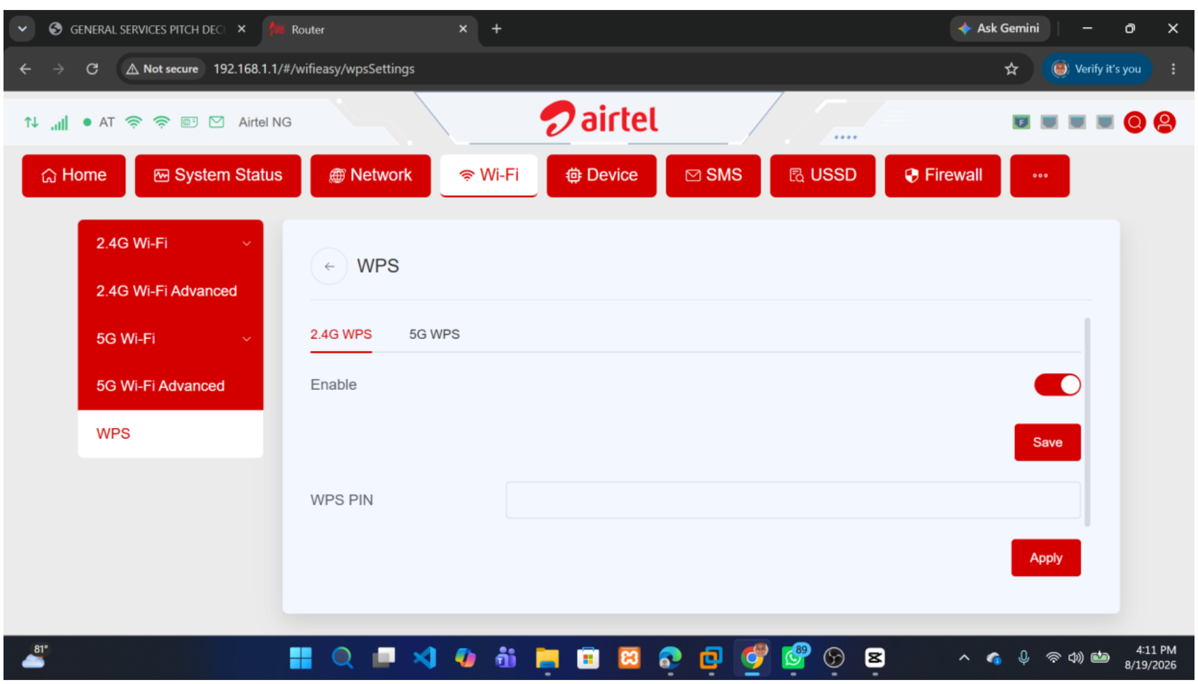


Phase-5-01-2.4GHz-Authentication.png

5 GHz authentication

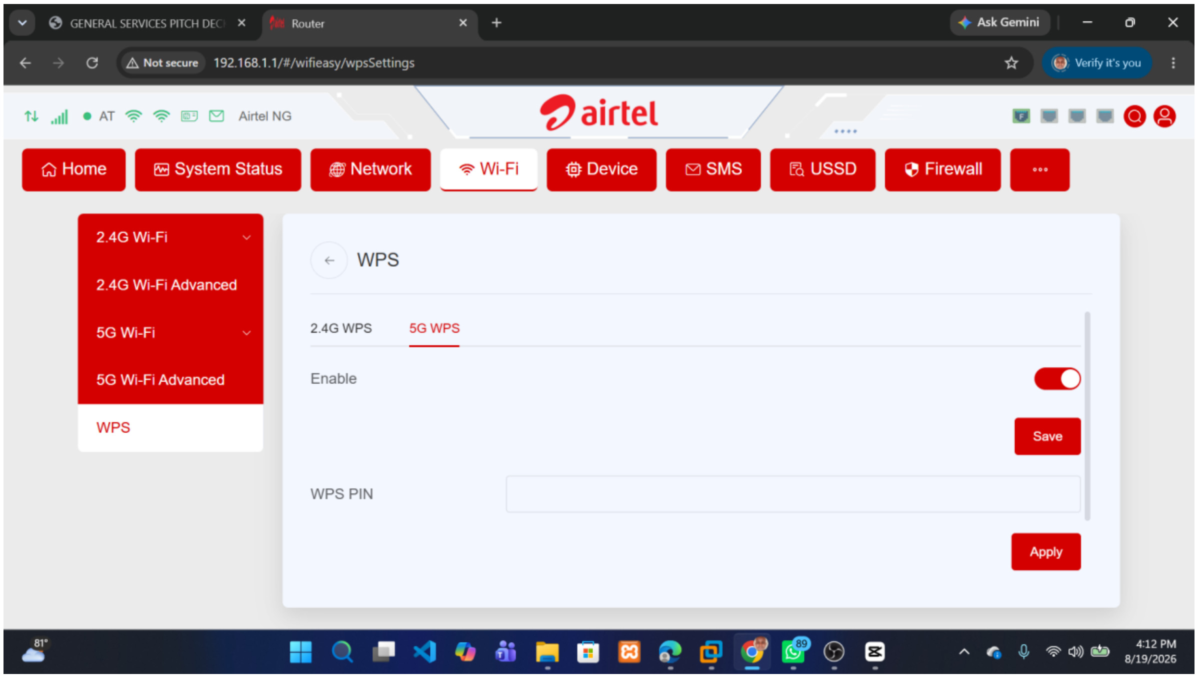
Phase-5-03-5GHz-Authentication.png

2.4 GHz WPS before



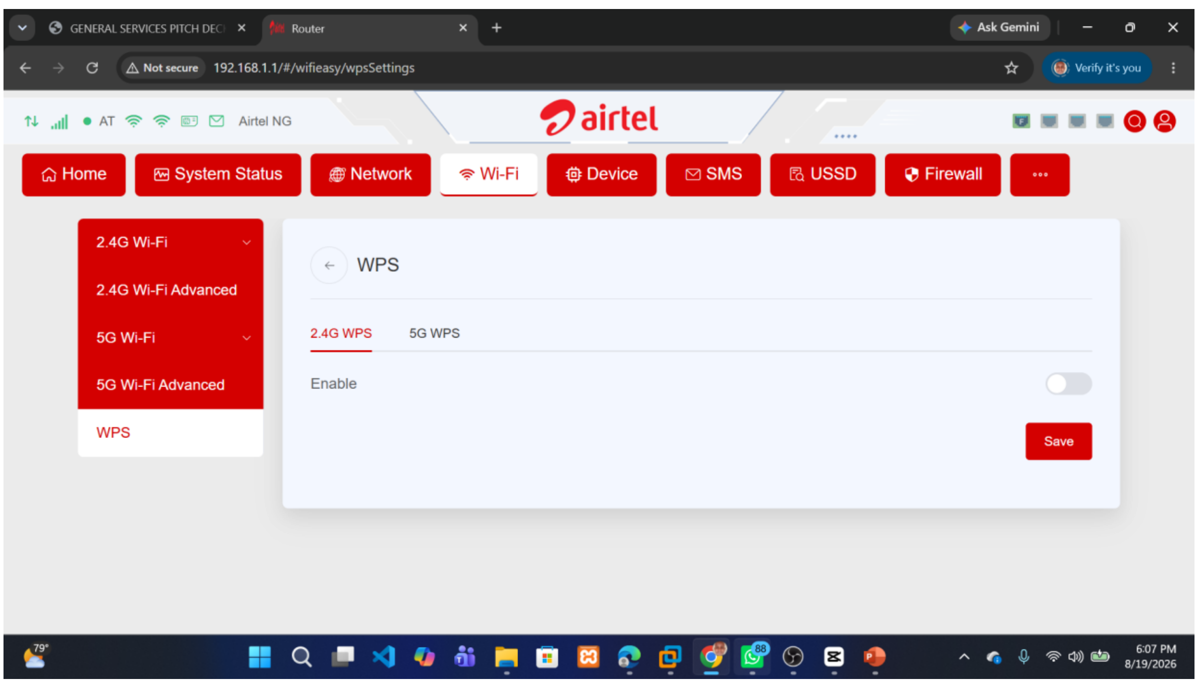
Phase-6-01-2.4GHz-WPS-Before.png

5 GHz WPS before



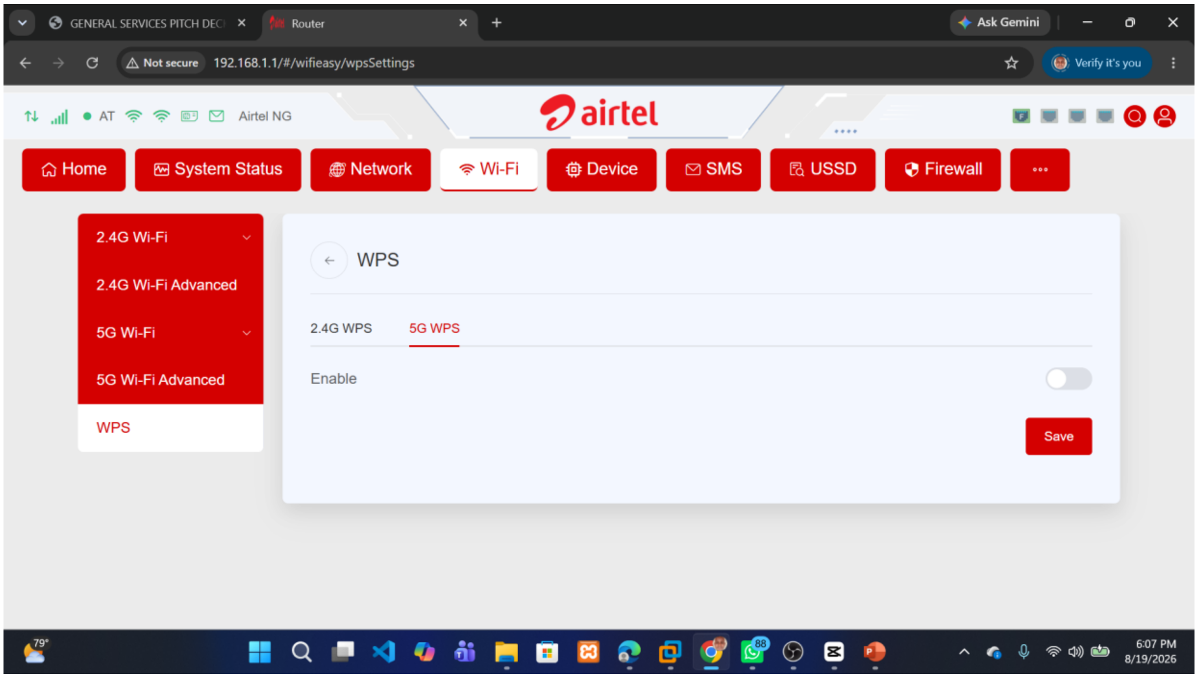
Phase-6-02-5GHz-WPS-Before.png

2.4 GHz WPS after



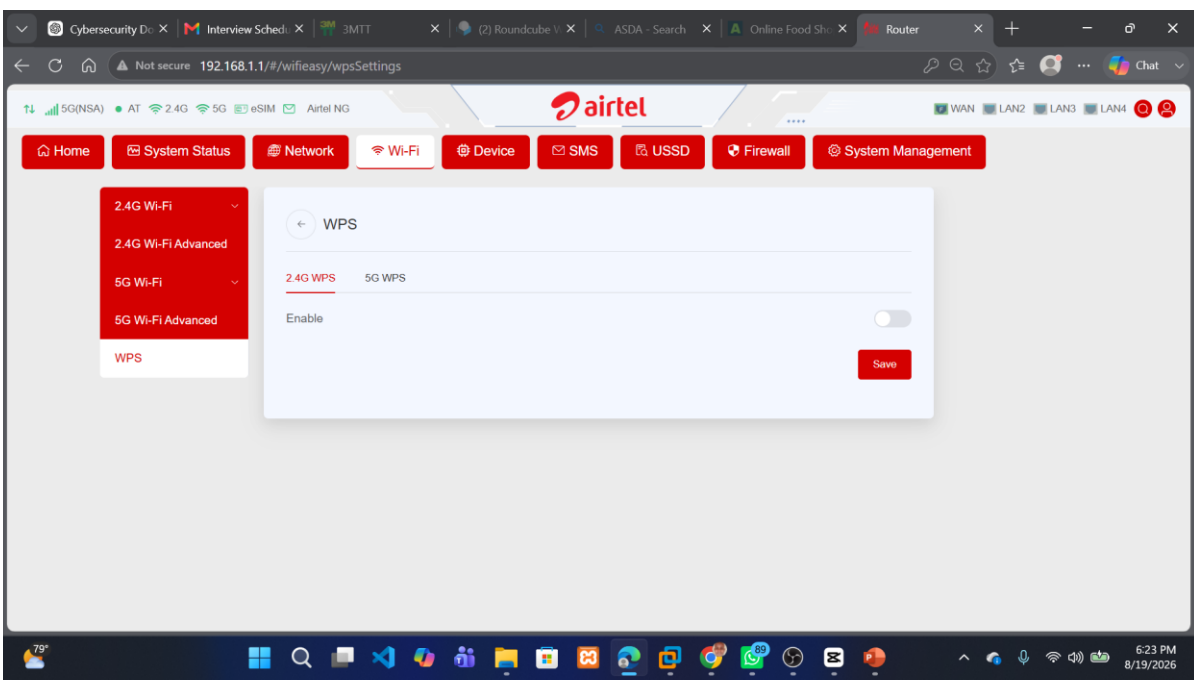
Phase-9-01-2.4GHz-WPS-After.png

5 GHz WPS after



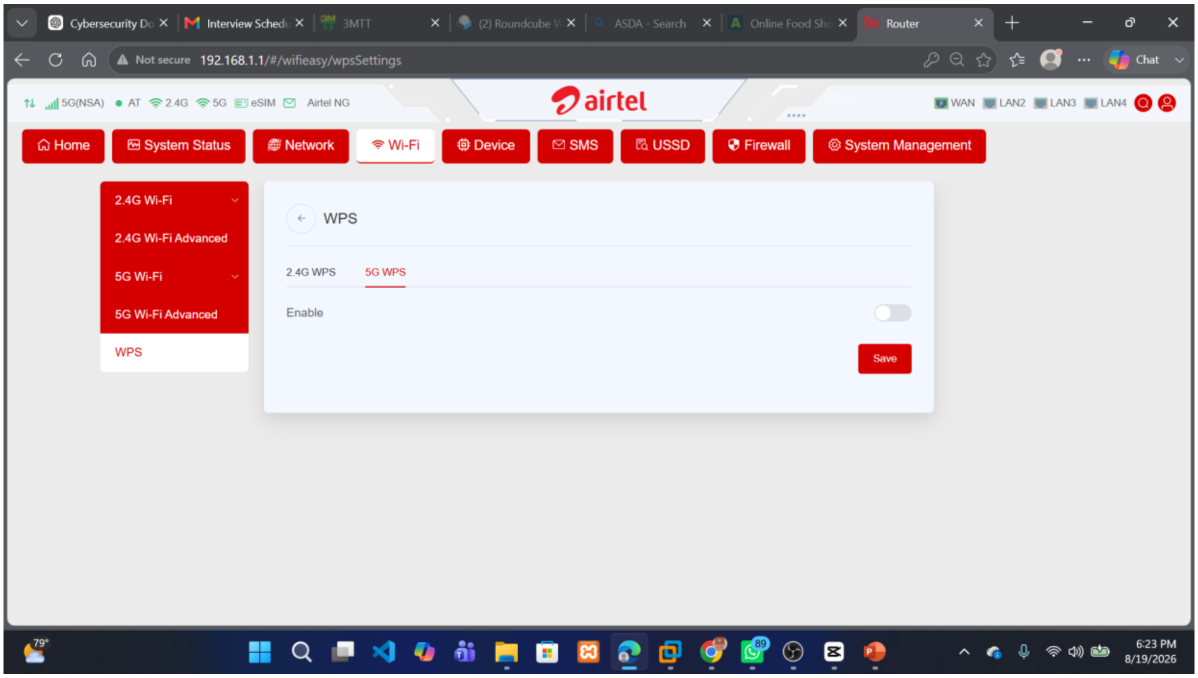
Phase-9-02-5GHz-WPS-After.png

2.4 GHz WPS verified



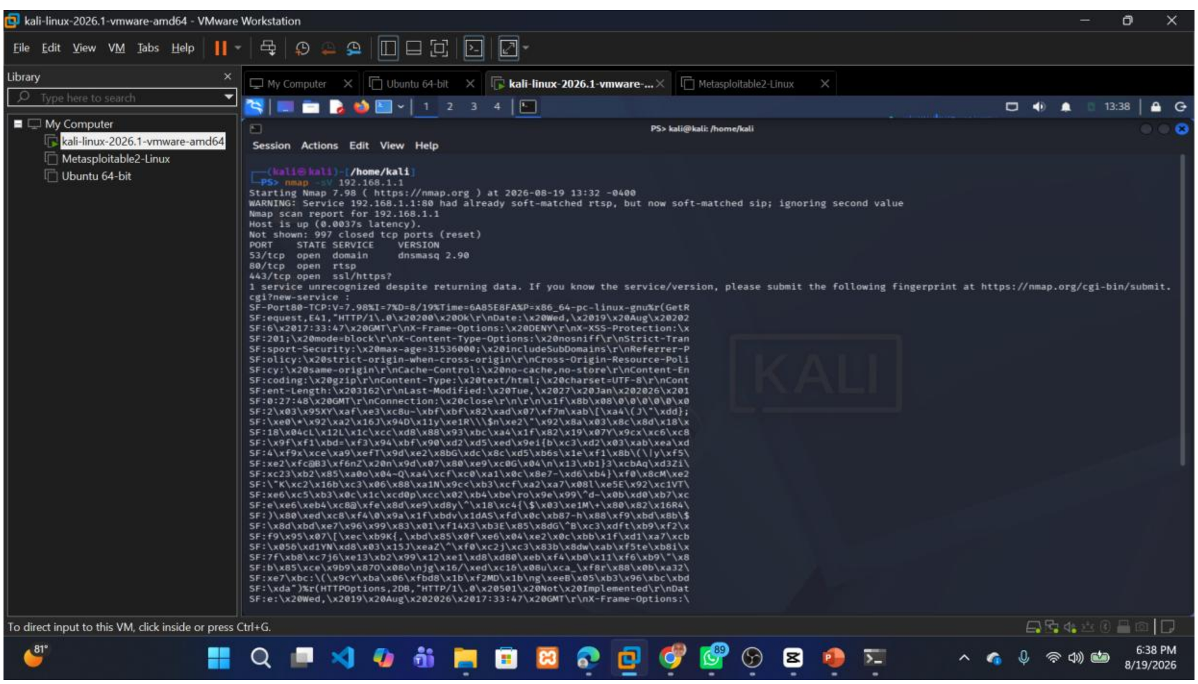
Phase-10-01-2.4GHz-WPS-Verified.png

5 GHz WPS verified



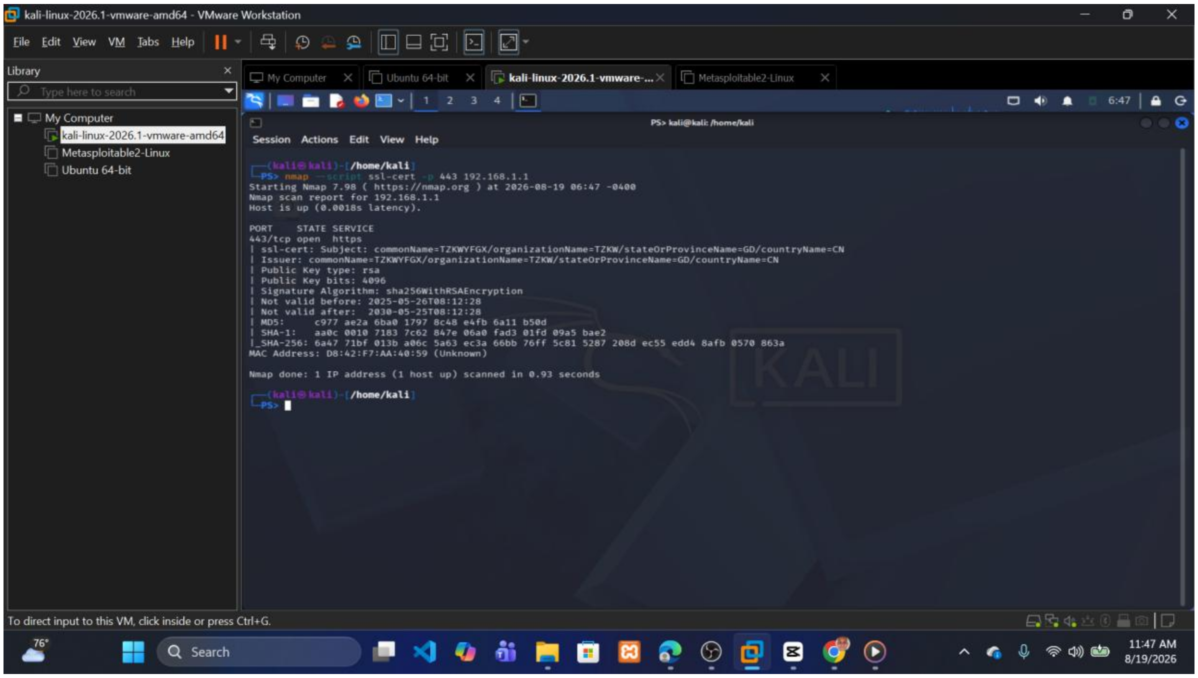
Phase-10-02-5GHz-WPS-Verified.png

Post-remediation Nmap



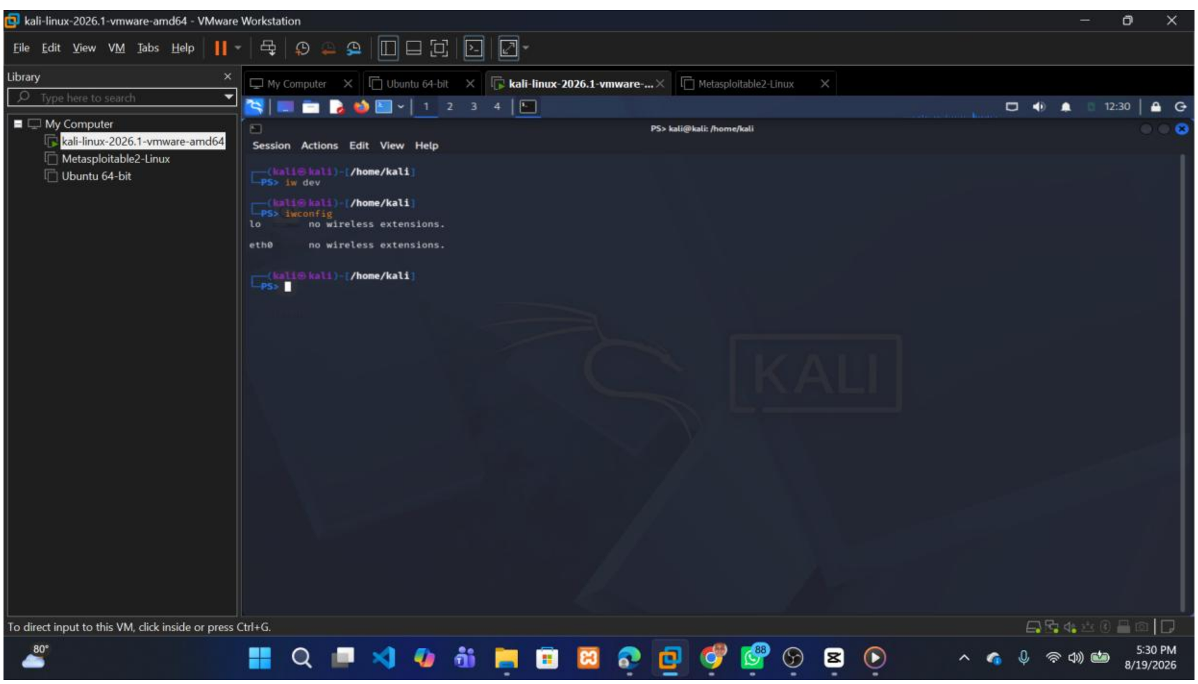
Phase-10-04-Post-Remediation-Nmap.png

TLS certificate



TLS Certificate.png

Kali wireless-interface limitation



Phase-7-04-IWCONFIG.png